

Reducing Cardiovascular Disease in Indonesia

Health Impact, Costs, and Economic Returns of Clinical and Public-Health Interventions, 2025–2050

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Reducing CVD in Indonesia

Why this matters for Indonesia

Cardiovascular disease is Indonesia's largest health burden

Under current trends, a person reaching age 30 faces roughly a **18.4% chance** of dying from cardiovascular disease before age 70 — the CVD “40q30”. Most of this is **preventable or treatable**.

- Premature CVD deaths (ages 30–69) have risen for two decades and remain high.
- Two complementary levers:
 - **Clinical care** — prevent and better treat established disease.
 - **Population prevention** — tobacco, salt, trans-fat, sugary drinks, alcohol.
- This analysis weighs their **health impact, cost, and economic return** together.

Source: Indonesia CVD model (calibrated to GBD 2023; UN World Population Prospects 2024). CVD 40q30 = probability of dying from cardiovascular disease between exact ages 30 and 70.

What this analysis set out to do

General aim

Estimate the **health and economic effects** of scaling up evidence-based **clinical** and **public-health** cardiovascular interventions in Indonesia through **2050**.

Aim 1 — Burden

Project the CVD burden and the premature-mortality indicator (CVD 40q30) under the **status quo** and under **alternative intervention scenarios**.

Aim 2 — Value for money

Estimate **deaths averted**, mortality reductions, **incremental costs**, and **cost-effectiveness** for each intervention and package.

Aim 3 — Economic return

Estimate **economic benefits**, **benefit-cost ratios**, and the **added value of combining** clinical and public-health action.

Horizon 2025–2050; the “status quo” baseline is the no-new-intervention comparator against which every scenario is measured.

Executive summary — the headline numbers

Combining clinical + public health is the big prize

Together they could avert about **6.2M deaths** over 2025–2050, cut the CVD 40q30 in 2050 by **31%**, and return about **79.4 int\$** in benefits per int\$ spent.

- **All clinical interventions:** $\approx 3.2\text{M}$ deaths averted; CVD 40q30 down 20% by 2050.
- **All public-health policies:** $\approx 3.3\text{M}$ deaths averted; benefit–cost ratio $\approx 1,290$.
- **Most impactful single levers:** **Primary CVD prevention** and **Tobacco tax**.
- **Affordability:** the combined package costs about US\$4,793 per death averted.

All figures from the model's cost/value workbooks. Deaths averted are cumulative 2025–2050 across modeled cardiovascular and diabetes causes; benefits valued per the Reference Case (Robinson 2019) in PPP international dollars (Robinson, Hammitt, and O'Keeffe 2019).

How the model works — in plain language

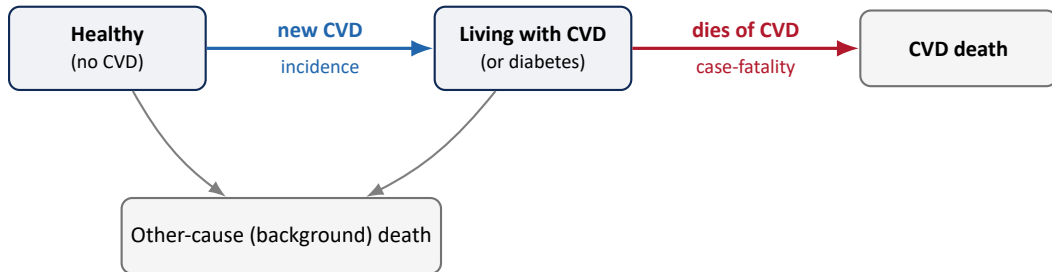
1. **People & demography.** Indonesia's population is projected by age and sex (UN World Population Prospects 2024).
2. **Starting health.** Disease and death rates are set to Indonesia's data (GBD 2023) and calibrated to match observed levels.
3. **Year-by-year simulation.** Each year people can develop cardiovascular disease or diabetes, survive, or die.
4. **Interventions.** Clinical and public-health scenarios change how many people develop disease and how many survive it.
5. **Outcomes.** The model reports deaths, healthy years (DALYs), premature-mortality (40q30), costs, cost-effectiveness, and economic returns.

One consistent engine

Every scenario runs through the **same** projection, so results are directly comparable and the **combined** package is modeled as one run — not a sum of separate estimates.

Calibration fits the model to GBD estimates using a standard numerical optimiser (Nelder–Mead). Life expectancy for lost-years and economic value uses UN WPP 2024.

How interventions change the disease pathway



Fewer people fall ill

Primary prevention & *all* public-health policies lower **incidence**.

More people survive

Secondary prevention, treatment & tobacco policy lower **case-fatality**.

Background mortality

Other-cause deaths from demography; unchanged by these interventions.

No recovery/remission transition is modeled: people living with CVD remain at elevated risk (a conservative assumption).

Source: model engine, 06_run_scenarios_indonesia_fair.R.

The interventions

Clinical interventions (FAIR Choices)

Intervention	Target population	Model effect
Primary CVD prevention	Adults at cardiovascular risk	Incidence
Type-2 diabetes treatment	People with type-2 diabetes	Case-fatality
Heart-attack secondary prevention	Heart-attack (IHD) survivors	Case-fatality
Stroke secondary prevention	Stroke survivors	Case-fatality
Chronic & acute heart-failure care	Heart-failure patients	Case-fatality
RHD secondary prevention & surgery	Rheumatic heart disease patients	Case-fatality

Implementation scenario (all clinical interventions): coverage scales up **linearly from 31% to 80%** of the eligible population between 2025 and 2050 (RHD surgery from a lower base). Effect sizes are drawn from the FAIR Choices intervention taxonomy; primary prevention lowers who develops CVD, while secondary prevention and treatment improve survival. Source: workbook `Selected_Interventions`.

Public-health interventions (fiscal & regulatory policy)

Policy	Risk factor	Model effect	Implementation lever
Tobacco tax	Tobacco	Incidence + survival	Raise tax to 75% of retail price
Tobacco policy package	Tobacco	Incidence + survival	Clean-air law, mass media, ad ban
Sodium policy package	Dietary salt/sodium	Incidence	Reformulation, labelling, media, procurement
Trans-fat elimination	Industrial trans-fat	Incidence	Best-practice industrial TFA ban
Sugary-drink (SSB) tax	Sugary drinks	Incidence	Price increase via excise
Alcohol tax / ad ban	Alcohol	Incidence	Price increase; advertising ban

Policies reduce exposure (smoking, salt, trans-fat, sugary-drink and alcohol intake), lowering CVD **incidence**. **Tobacco** policies additionally improve **survival** after CVD. Packages are modeled as single joint runs of their component measures. Source: workbook `Selected_Interventions / Policy_Lever`s.

Data sources

Model component	Primary source	Years / version	Use in model
Population & demography	UN World Population Prospects	2024 revision	Population, background mortality, life expectancy
Cause-specific mortality & disease rates	GBD (Indonesia)	2023	Baseline burden & calibration targets
Disability weights (healthy-years lost)	GBD	2023	DALYs
Clinical intervention effects	FAIR Choices intervention taxonomy	current	Incidence & case-fatality effects
Public-health effect sizes	Published risk-factor evidence (in workbook)	current	Incidence & survival effects
Intervention & policy costs	Programme & policy unit costs (in workbook)	price year 2023	Budget impact & cost-effectiveness
Economic valuation (VSL/VSLY)	Reference Case guidance (Robinson 2019)	2019	Monetized benefits & benefit–cost
International 40q30 comparison	WPP-scaled ASEAN comparison series	1995–2050	Benchmarking Indonesia vs. neighbour

Sources verified in the modeling pipeline and input workbooks. Specific effect-size and cost citations are documented in the input workbooks. Population: UN World Population Prospects 2024; GBD (GBD 2023 Cardiovascular Diseases and Risks Collaborators 2025; Schumacher et al. 2024); economic valuation (Robinson, Hammitt, and O’Keeffe 2019).

Health impact and value for money

Public health: impact by policy

Public-health policy	Deaths averted (cumulative)	CVD 40q30 2050	40q30 reduction 2050	Deaths reduction 2050
Tobacco tax	1,610,000	17.3%	6.2%	4.4%
Sodium policy package	927,000	17.7%	4.0%	2.9%
Tobacco policy package	727,000	17.9%	2.8%	2.0%
Trans-fat elimination	144,000	18.3%	0.7%	0.5%
Sugary-drink tax	43,000	18.4%	0.2%	0.1%
Alcohol advertising ban	5,000	18.4%	0.0%	0.0%
Alcohol tax	4,000	18.4%	0.0%	0.0%
All public health (combined)	3,328,000	16.0%	13.3%	9.7%

Cumulative deaths averted over 2025–2050 vs. the status-quo baseline. Packages (tobacco, sodium) are single joint runs of their component measures; individual sodium/tobacco measures are listed in the appendix. Source: public-health cost/value workbook.

Clinical: impact and cost-effectiveness

Clinical intervention	Deaths averted (cumulative)	CVD 40q30 2050	40q30 red. 2050	Deaths red. 2050	Cost per death averted	Annual cost per capita
Primary CVD prevention	2,291,000	15.2%	17.4%	11.2%	US\$521	US\$0.24
Type-2 diabetes treatment	491,000	18.4%	0.3%	2.1%	US\$29,566	US\$2.99
Heart-attack secondary prevention	231,000	18.1%	1.6%	0.8%	US\$20,439	US\$0.97
Chronic heart-failure care	97,000	18.3%	0.6%	0.3%	US\$7,835	US\$0.16
Acute heart-failure care	84,000	18.4%	0.4%	0.3%	US\$14,928	US\$0.26
Stroke secondary prevention	52,000	18.4%	0.3%	0.2%	US\$53,006	US\$0.57
RHD secondary prevention	11,000	18.4%	0.0%	0.0%	US\$15,479	US\$0.03
RHD cardiac surgery	2,000	18.4%	0.0%	0.0%	US\$3,059,822	US\$1.12
All clinical (combined)	3,202,000	14.7%	20.1%	14.6%	US\$9,190	US\$6.02

Cost per death averted uses **discounted** incremental cost (3%/yr). Annual cost per capita = undiscounted incremental cost averaged over 26 years and Indonesia's population. Costs in market US\$ (price year 2023); perspective: health system. Source: clinical cost/value workbook.

Clinical: benefit–cost analysis

Clinical intervention	Deaths averted	Cost per death averted	Benefits (PV)	Net benefit (PV)	Benefit–cost ratio
Primary CVD prevention	2,291,000	US\$521	Int\$2.6T	Int\$2.6T	738
Type-2 diabetes treatment	491,000	US\$29,566	Int\$562.1B	Int\$518.6B	12.9
Heart-attack secondary prevention	231,000	US\$20,439	Int\$263.3B	Int\$249.2B	18.6
Chronic heart-failure care	97,000	US\$7,835	Int\$110.1B	Int\$107.8B	48.4
Acute heart-failure care	84,000	US\$14,928	Int\$96.2B	Int\$92.4B	25.4
Stroke secondary prevention	52,000	US\$53,006	Int\$59.0B	Int\$50.7B	7.2
RHD secondary prevention	11,000	US\$15,479	Int\$12.5B	Int\$12.0B	24.5
RHD cardiac surgery	2,000	US\$3,059,822	Int\$2.1B	Int\$-14.5B	0.1
All clinical (combined)	3,202,000	US\$9,190	Int\$3.7T	Int\$3.6T	41.7

Benefits = value of averted mortality (preferred VSL, income elasticity 1.5), in PPP international dollars, base year 2025, discounted at 3%/yr; standing: societal; scope: partial (mortality benefit only). A ratio above 1 means benefits exceed costs. Source: clinical cost/value workbook, preferred-VSL case.

Public health: cost-effectiveness

Public-health policy	Deaths averted (cumulative)	CVD 40q30 2050	40q30 red. 2050	Deaths red. 2050	Cost per death averted	Annual cost per capita
Tobacco tax	1,610,000	17.3%	6.2%	4.4%	US\$126	US\$0.04
Sodium policy package	927,000	17.7%	4.0%	2.9%	US\$192	US\$0.03
Tobacco policy package	727,000	17.9%	2.8%	2.0%	US\$161	US\$0.02
Trans-fat elimination	144,000	18.3%	0.7%	0.5%	US\$306	US\$0.01
Sugary-drink tax	43,000	18.4%	0.2%	0.1%	US\$4,708	US\$0.04
Alcohol advertising ban	5,000	18.4%	0.0%	0.0%	US\$6,099	US\$0.01
Alcohol tax	4,000	18.4%	0.0%	0.0%	US\$47,840	US\$0.04
All public health (combined)	3,328,000	16.0%	13.3%	9.7%	US\$295	US\$0.18

Fiscal and regulatory policies reach the whole population at very low per-person cost, so cost per death averted is far lower than for clinical care. Same costing conventions as the clinical table. Source: public-health cost/value workbook.

Public health: benefit–cost analysis

Public-health policy	Deaths averted	Cost per death averted	Benefits (PV)	Net benefit (PV)	Benefit–cost ratio
Tobacco tax	1,610,000	US\$126	Int\$1.8T	Int\$1.8T	2,990
Sodium policy package	927,000	US\$192	Int\$1.1T	Int\$1.1T	1,980
Tobacco policy package	727,000	US\$161	Int\$825.5B	Int\$825.1B	2,340
Trans-fat elimination	144,000	US\$306	Int\$165.4B	Int\$165.3B	1,250
Sugary-drink tax	43,000	US\$4,708	Int\$49.3B	Int\$48.7B	81.0
Alcohol advertising ban	5,000	US\$6,099	Int\$5.3B	Int\$5.2B	62.3
Alcohol tax	4,000	US\$47,840	Int\$4.8B	Int\$4.2B	7.9
All public health (combined)	3,328,000	US\$295	Int\$3.8T	Int\$3.8T	1,290

Because fiscal/regulatory policies cost little to implement relative to the mortality they avert, their benefit–cost ratios are very large. Same valuation conventions (preferred VSL, PPP_international_dollars, base year 2025, 3%/yr) as the clinical BCA. Source: public-health cost/value workbook.

The combined package

Clinical + public health: better together

Package	Deaths averted (cumulative)	40q30 red. 2050	CVD 40q30 2050	Total cost (PV)	Cost per death averted	Benefit–cost ratio
All clinical (combined)	3,202,000	20.1%	14.7%	US\$29.4B	US\$9,190	41.7
All public health (combined)	3,328,000	13.3%	16.0%	US\$980.2M	US\$295	1,290
Clinical + public health (combined)	6,181,000	30.6%	12.8%	US\$29.6B	US\$4,793	79.4

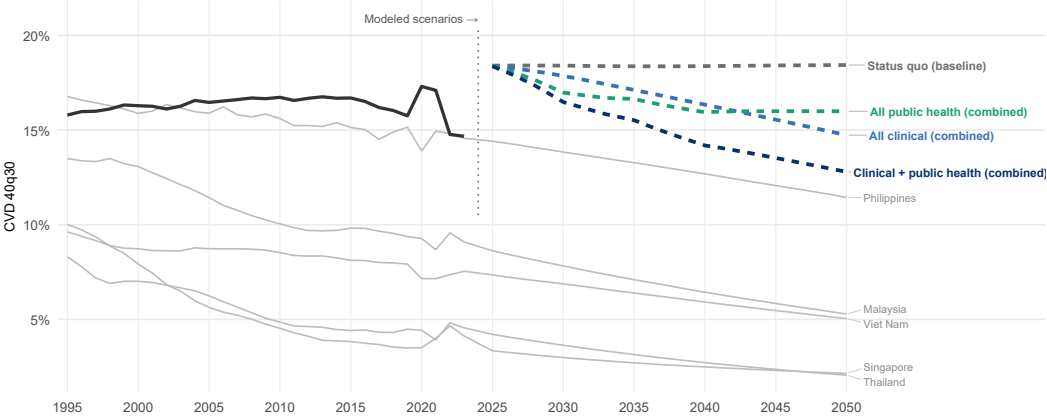
The combined package is modeled as one run — not a sum

Because clinical and public-health effects overlap, the joint scenario is projected in a single run. It averts about **6.2M deaths**, brings the 2050 CVD 40q30 down to **12.8%** (from 18.4% at baseline), and yields a benefit–cost ratio of about **79.4** — a net benefit of roughly **Int\$7.0T**.

CVD premature mortality: trajectory and neighbours

CVD 40q30 in Indonesia: observed, projected, and neighbours

Probability of dying from CVD between ages 30 and 70



Indonesia observed history (solid, to 2023) and ASEAN comparators are WPP-scaled reference series.
Modeled Indonesia scenarios (dashed, 2025-2050) use the GBD-calibrated model baseline, which sits somewhat higher.

What it means for Indonesia

Policy implications

- **Biggest health gains come from doing both.** The combined package averts far more (~6.2M) than either family alone.
- **Public-health policies are exceptional value.** Population-wide fiscal/regulatory measures avert deaths at a fraction of clinical cost (ratios in the thousands).
- **Clinical scale-up delivers large, durable gains** in survival and premature mortality, at conventional cost-effectiveness.
- **Sequence for early wins.** Tobacco taxation and other prevention policies can start cutting risk quickly and cheaply, while clinical capacity scales up.
- **The 40q30 target is within reach.** The combined package bends Indonesia's premature-CVD-mortality curve down by about 31% by 2050.

These are model findings; translating them into policy requires judgement on feasibility, financing and delivery. Model results quantify the *potential*; they are not a delivery guarantee.

Limitations

- Results depend on **modeled epidemiological estimates** (GBD) and calibration; real burden may differ.
- **Effect sizes, future coverage, and unit costs are uncertain**; interventions are assumed to reach and sustain their targets.
- The disease model is **simplified** (healthy → living-with-CVD → death; no remission; no comorbidity interactions).
- Economic returns value **averted mortality only** (a partial benefit-cost scope) — morbidity, productivity and downstream cost offsets are omitted, so returns are, if anything, understated.
- Projections are **scenarios, not forecasts or guarantees**; they show what is achievable if interventions are delivered as modeled.

Three things to remember

1. Both, together

Clinical + public health averts **~6.2M deaths** by 2050 — more than either alone.

2. Prevention pays

Population policies return on the order of **1,290×** their cost.

3. A reachable target

Premature CVD mortality (40q30) falls **31%** by 2050 under the combined package.

Combined action against CVD could avert about **6.2M deaths** and return about **79.4 int\$ per int\$ invested** through 2050.

Technical appendix

Appendix 1 — Scenario crosswalk

Family	Workbook scenario id	Executive label
Clinical (FAIR Choices)	I_CVD_PRIMARY	Primary CVD prevention
	I_T2D_TREATMENT	Type-2 diabetes treatment
	I_IHD_SECONDARY	Heart-attack secondary prevention
	I_STROKE_SECONDARY	Stroke secondary prevention
	I_HF_CHRONIC	Chronic heart-failure care
	I_HF_ACUTE	Acute heart-failure care
	I_RHD_SECONDARY	RHD secondary prevention
	I_RHD_SURGERY	RHD cardiac surgery
	all	All clinical (combined)
	I_PH_TOBACCO_TAX	Tobacco tax
Public health	I_PH_TOBACCO_POLICY	Tobacco policy package (clean-air+media+ad-ban)
	I_PH_SALT_POLICY	Sodium policy package (reformulation+labelling+media+environment)
	I_PH_TFA_POLICY	Trans-fat elimination
	I_PH_SSB_TAX	Sugary-drink tax
	I_PH_ALCOHOL_TAX	Alcohol tax
	I_PH_ALCOHOL_POLICY	Alcohol advertising ban
	all_public_health	All public health (combined)
Combined	all_clinical_public_health	Clinical + public health (combined)

Packages are single joint runs of their child measures (shown in parentheses); child measures are omitted from the main tables to avoid double-counting.

Appendix 2 — Definitions, horizons and assumptions

Metrics

- **Deaths averted (cumulative)**
$$= \sum_{t=2025}^{2050} (\text{baseline deaths}_t - \text{scenario deaths}_t),$$

across modeled CVD & diabetes causes.
- **% reduction in deaths, 2050**
$$= 100 \times (\text{baseline} - \text{scenario}) / \text{baseline in 2050}.$$
- **CVD 40q30** = probability of dying of CVD between exact ages 30 and 70 (six CVD causes, both sexes, single-year life table).
- **% reduction in 40q30, 2050**: analogous to deaths.

Model. Discrete-time state-transition model; single-year ages 0–95; six CVD causes (IHD, ischemic stroke, intracerebral haemorrhage, hypertensive heart disease, rheumatic heart disease, cardiomyopathy & myocarditis) plus type-2 diabetes; calibrated to GBD 2023 (Nelder–Mead); demography and life expectancy from UN WPP 2024. **Validation:** Baseline reconciled across all three workbooks (2050 deaths agree to 0.0000%; 2050 CVD 40q30 to 0.0000%). All economic values were recalculated live from the formula workbooks.

Costing & valuation

- Horizon 2025–2050; discount rate 3%/yr; base year 2025; price year 2023.
- Costs in market US\$; cost perspective: health system.
- Benefits: averted-mortality value, **preferred VSL** (income elasticity 1.5), in PPP international dollars; standing societal; scope: partial mortality benefit.
- **Cost per death averted** uses discounted incremental cost.

Appendix 3 — References

- GBD 2023 Cardiovascular Diseases and Risks Collaborators. 2025. “Global, Regional, and National Burden of Cardiovascular Diseases and Risk Factors in 204 Countries and Territories, 1990–2023.” *Journal of the American College of Cardiology* 86 (22): 2167–2243.
- Robinson, Lisa A., James K. Hammitt, and Lucy O’Keeffe. 2019. “Valuing Mortality Risk Reductions in Global Benefit-Cost Analysis.” *Journal of Benefit-Cost Analysis* 10 (S1): 15–50. <https://doi.org/10.1017/bca.2018.26>.
- Schumacher, Austin E., Hmwe Hmwe Kyu, Catherine M. Antony, Aleksandr Y. Aravkin, and Gulrez. 2024. “Global Age-Sex-Specific Mortality, Life Expectancy, and Population Estimates in 204 Countries and Territories and 811 Subnational Locations, 1950–2021, and the Impact of the COVID-19 Pandemic: A Comprehensive Demographic Analysis for the Global Burden of Disease Study 2021.” *The Lancet* 403 (May): 1989–2056. [https://doi.org/10.1016/S0140-6736\(24\)00476-8/ATTACHMENT/B6671C4E-CF06-47EB-AC9B-6A42B327599D/MMC3.PDF](https://doi.org/10.1016/S0140-6736(24)00476-8/ATTACHMENT/B6671C4E-CF06-47EB-AC9B-6A42B327599D/MMC3.PDF).